

Manikomevwe Udjoh

Date: 11/15/23

Westmorland River Watershed Project

Identify the watershed site you have chosen (see Final Project Scenarios document in SAM):

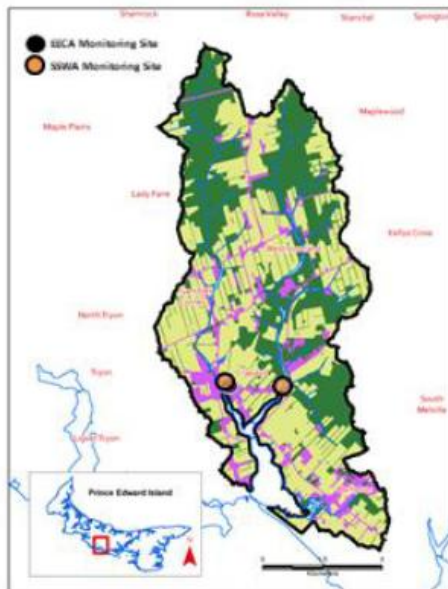
Watershed Name: Westmorland River Watershed

Location: Prince Edward Island

Areas of concern: sediment laden run-off (red water) events occur frequently. Elevated summer water temperature. Low dissolved oxygen concentrations

Relevant Watershed Stats (area, land use, human resources, budget, photos, maps, etc.):

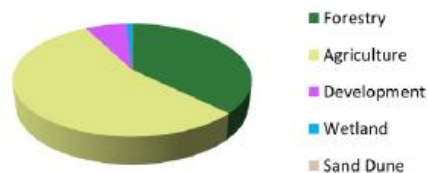
**Westmoreland River
Map**



Water Quality Score



Land Use



Watershed Area
44.61 km²

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Source: [2018 water quality report cards final for web 07072020.pdf \(princeedwardisland.ca\)](#)

Area: 4,461Ha (44.61Km²)

Land Use:

Agriculture – 60%

Development – 7%

Forest – 30%

Sand Dune – 1%

Wetland – 2%

Refer to your summary table in your Learner Workbook. Fill in the table to identify your considerations about which technology tools would be most and least appropriate for your watershed site scenario:

	Resolution & Accuracy – <i>Do you need high resolution imagery? Do you require accurately georeferenced data?</i>	Coverage Area – <i>How large is the area you are looking to cover?</i>	Budget & Financial – <i>What financial constraints are in place?</i>	Timing of Data – <i>Do you require up to date information or is previously collected data suitable?</i>	Data Sources & Storage – <i>Where will you source the data? Do you need to collect it yourself?</i>	Outputs Required – <i>What outputs are required to achieve your goals? (e.g., Orthomosaic, digital surface model, slope map, measurement, etc.)</i>	Analysis Software – <i>What software will be required to work with your data?</i>	Additional Considerations – <i>Is there anything else you will need to consider? (e.g., environment, weather, IT, human resources, etc.)</i>
Project Requirements	10m resolution Satellite data and 1m DTM. Already accurately Georeferenced	4,461Ha	Free – LiDAR data and GIS software will be used from the Government of Canada website.	Both- It will be essential to make comparisons by knowing what the previous situation was and what exists currently.	Open Maps (Government of Canada)	A map showing slope, buffer within 250m of streams within the Westmoreland River watershed	QGIS	Watershed sensitization

The Westmoreland watershed is located in the south-central portion of Prince Edward Island, running from the Victoria/Hampton area in the south, through the communities of Crapaud and Westmoreland, north to the area between Rose Valley and Maplewood. The area covers almost 5000 hectares, and the main areas of concern are water quality – high nitrate levels and sediment laden run-off (red water) occurs frequently. There also exists the occurrence of elevated summer water temperature and low dissolved oxygen concentrations, so it would be necessary to understand stream flow, flooding and erosion occurrence and the slope to highlight how water flows over land and the highest points (elevation) of the land. For this case study, LiDAR and/or satellite imagery have been identified as the most appropriate technology for collecting the relevant data to support watershed resilience in the Westmorland River Watershed. LiDAR has the capability to penetrate through vegetation, offering precise elevation data

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from the Earth's surface. It delivers a constant stream of elevation information for expansive regions, enabling the modeling of waterflow, evaluating erosion risk, and devising strategies to address potential environmental concerns arising from soil erosion.

How would you use GIS software to generate your required outputs from the data you source?

Terrain Modeling & Data Processing: Using the PEI watershed boundaries dataset, I obtained the shapefile for the Westmoreland watershed and placed it against the satellite base map. This gave me an idea of the location of my study area on the ground. I utilized the Open Canada website to download processed LIDAR data in the form of a Digital Terrain Model (DTM), I built a raster by using the geoprocessing tool to clip the watershed shapefile against the LIDAR data. This depicts the earth's surface as well as elevation information. Next, I calculated the slope of the watershed using the slope tool to highlight the high and low points of the watershed. This helps to identify areas on the land that are erosion prone such as low lying areas and bare soil. The map output below depicts the Westmoreland River Watershed Slope.

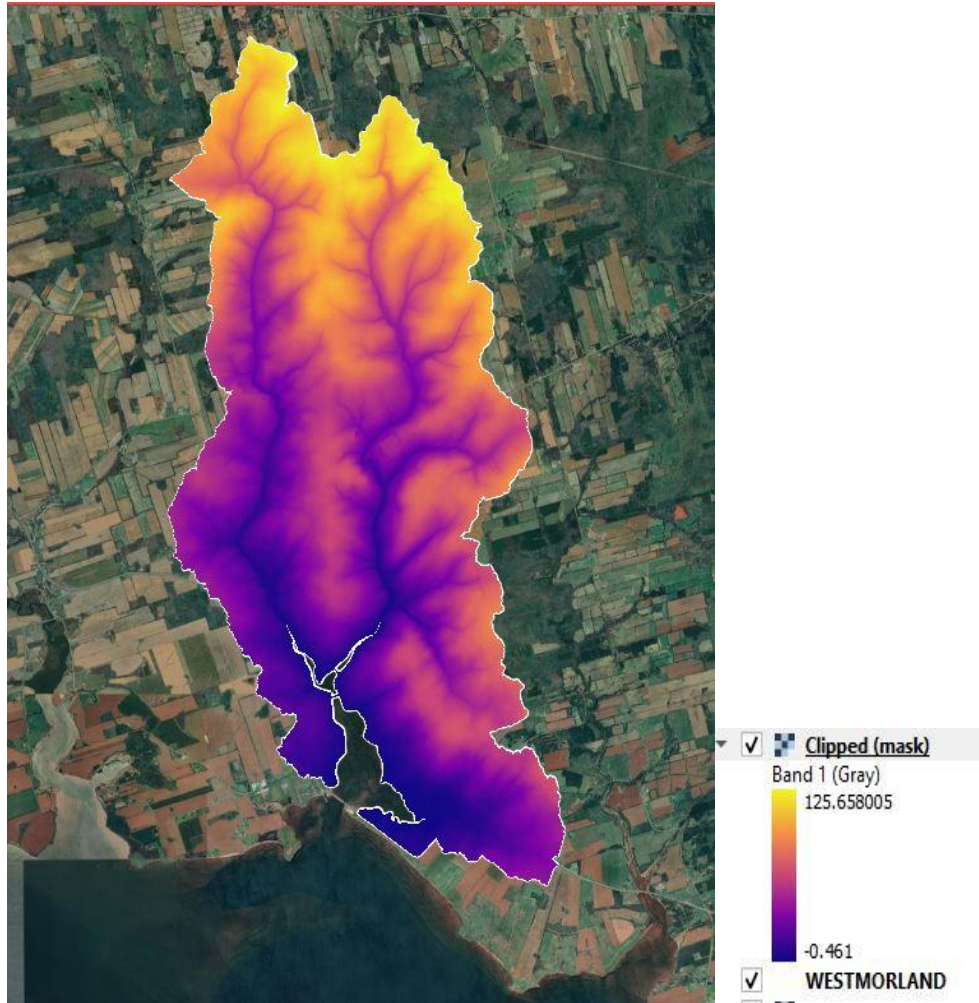


Fig1: Map of Westmoreland watershed depicting the slope.

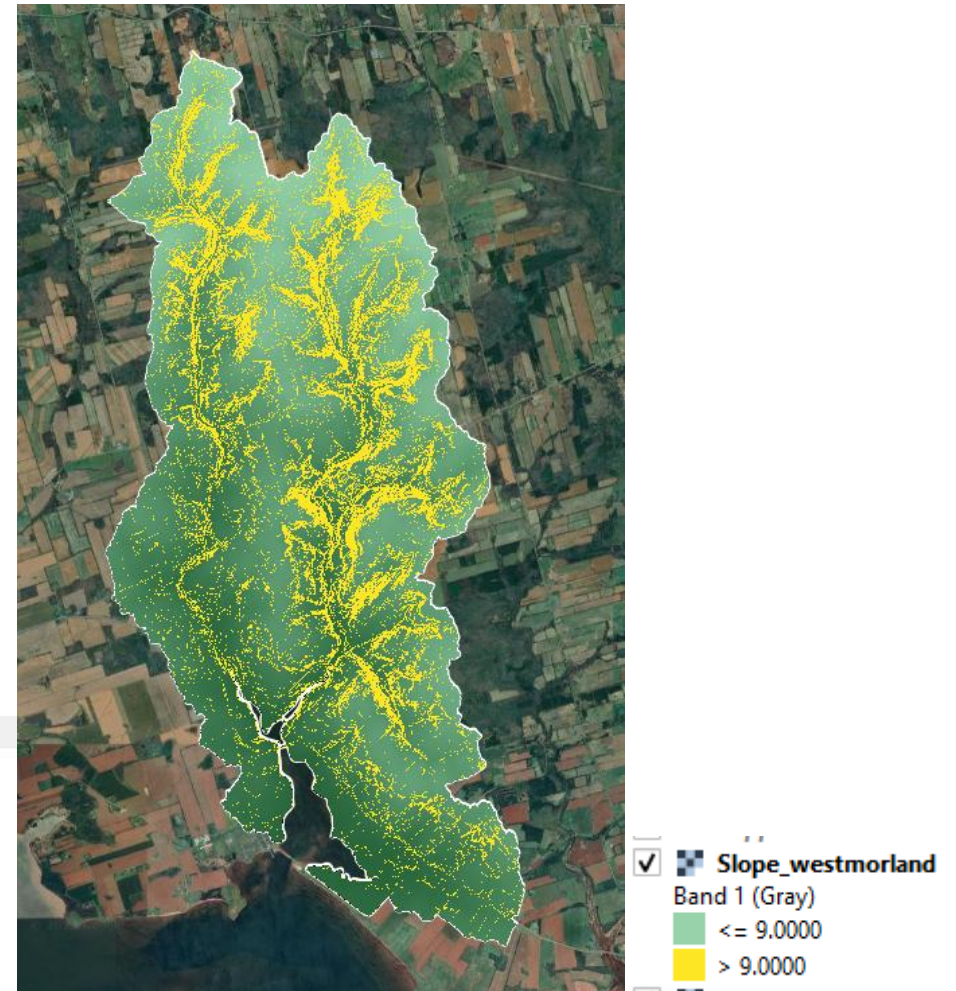


Fig2: Map of Westmoreland watershed showing the classified slope.

Hydrological Analysis & Erosion Risk Assessment:

Using the elevation data as well as the hydro-network data. The stream network coupled with areas of high slope which are also near streams can help to identify in the Springtime, areas of massive soil loss as well as where sediments flow into the water bodies. This also helps to identify flow paths and modeling waterflow patterns. Figure 3 below shows the output map, showing the hydro network added to the watershed map, showing potential hotspots and areas prone to erosion.

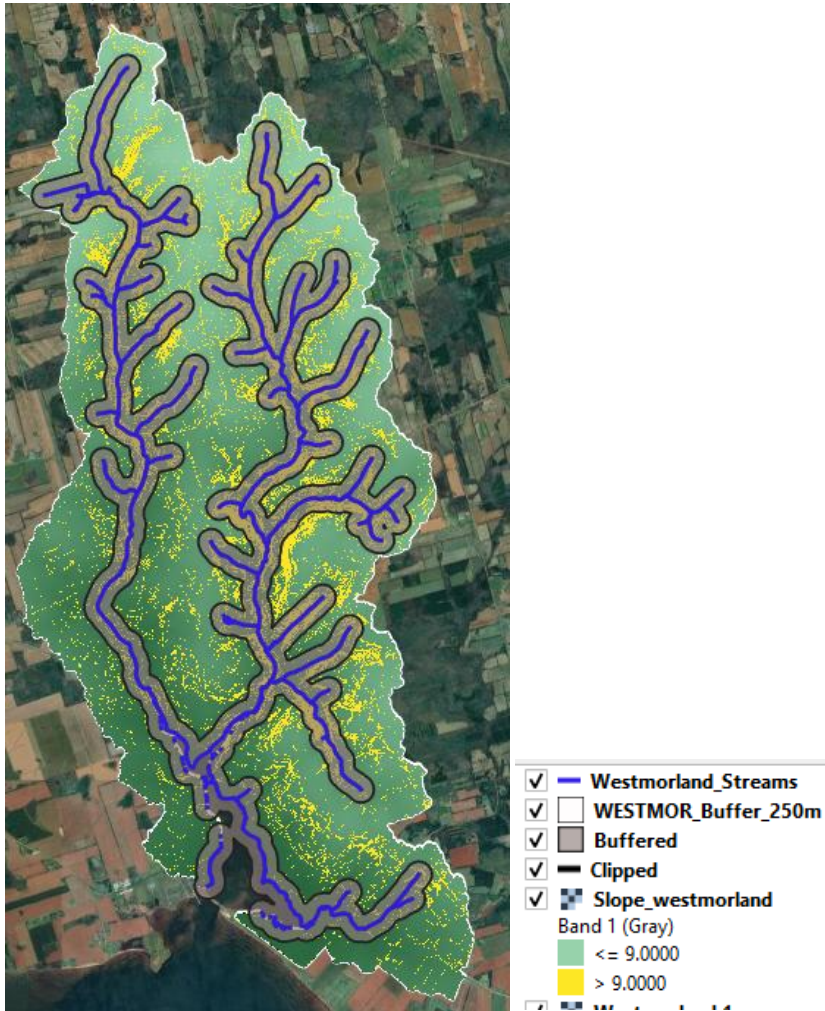


Fig3:Westmoreland watershed showing slope and potential areas prone to erosion.



Fig4: Dissolved Ground Cover calculation for Westmoreland watershed.

I also made use of satellite imagery. Since the watershed is prone to erosion and data from different periods of time need to be analyzed, satellite imagery is very important in conducting the analysis to calculate the percentage of the watershed that has ground cover. This is one advantage of utilizing satellite imagery, it can be obtained for different periods in time. It is important to know just how much ground cover is available on the watershed so we can tell exactly what area of land will be open to run off. Utilizing “Corpenicus.eu”, I panned to PEI and identified my area of interest by drawing a rectangle over the Westmoreland watershed region and used the Sentinel 2 mission data for the date November 19, 2022. I downloaded the data for Red, Near InfraRed and Truecolor and clipped them to the Westmoreland watershed shapefile . Near Infrared is very valuable for picking up what we cannot see with the naked eyes such as the level of chlorophyll in

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vegetation and ascertain the health of vegetation and for calculating the Normalized Difference Vegetative Index (NDVI). Higher NDVI values mean greener areas and lower values mean more bare soil. Using the Raster calculator tool, I calculated the NDVI of the watershed using the formula $\frac{\text{NIR-Red band}}{\text{NIR band} + \text{Red band}}$. Please see the results of the NDVI calculation below. The results of the Cover calculation were obtained using the Field Calculator and the area covered came to 3790 Ha which amounts to 85% cover.

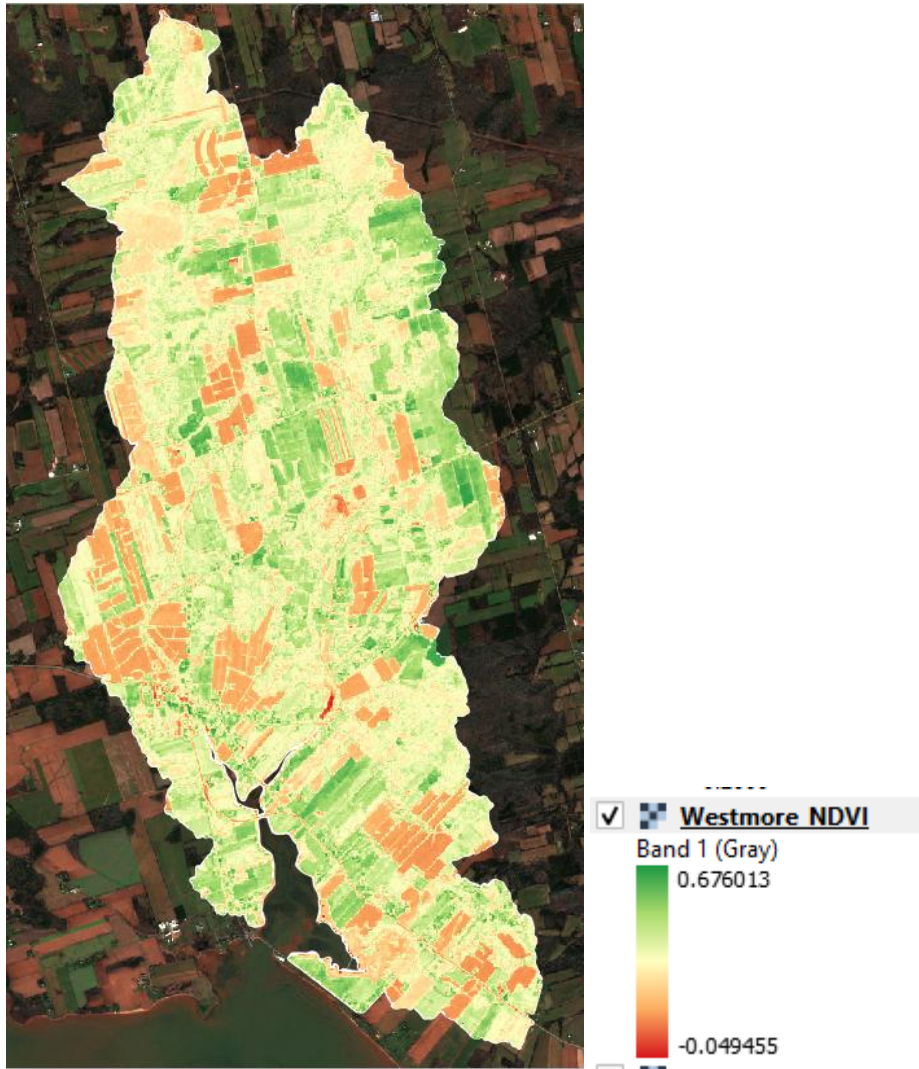


Fig5: NDVI calculation for Westmoreland watershed.

Why would this information be valuable for the Wheatley River Improvement Group?

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Documentation and Reporting:

This information can be used to develop plans to mitigate potential environmental issues related to soil erosion. The information will help in visualizing the landscape and Environmental Planning in identifying sensitive areas and proposing appropriate land management strategies.

From research, I found out that Island streams are known to be predominately groundwater during low flow, which means that nitrate results from data collected during the summer months (June – September) is used. To account for inter-annual variation at least 2 – 3 years of sampling results is required, and this should be from the current year and at least the two previous years. This is another reason why information from previous periods and sensitization is necessary. Organizations like the Watershed group help to assess results of previous targeted efforts (i.e.. Ensuring grass outlets are present, haybale installation, silt dams, etc.) to mitigate potential issues in areas where there is high slope combined with bare soil.